

Generalization & overview

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Research Design for Social Sciences
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Roadmap

Review

Generalizing results

Complementary tests

Extra: Synthetic control methods

Questions

Causal methods again, in detail

Re-cap & example

Questions & review

- Questions?

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- Key points from last year experience:
 - Need to think in a **broader** way
 - Do not stay fixated on a method
 - There's never a single solution/strategy

Questions & review

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- Key points from last year experience:
 - Need to think in a **broader** way
 - Do not stay fixated on a method
 - There's never a single solution/strategy
- Key thing I'd like you to take from this course:
learn how to **plan**, execution will come later

Design example

Let's say we're interested in exploring the problem of
AI automation and the labor market,
where do we start?

Remember:

1. Topic
2. RQ
3. Theory and mechanisms
4. Concepts, measurement
5. Strategy
6. Results, interpretation & relevance

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Generalizing results

- The **Credibility Revolution** in social sciences
 - Internally valid studies, context-specific
 - Accumulating generalizable knowledge by repetition
 - 'Valid' studies and synthesis studies
 - Practice-oriented, Esther Duflo: 'The Economist as Plumber'
- What are we really learning? How much *knowledge* we gain?

Generalizing results

1. Connection theory-empirics

- We start with big questions and then narrow them down
- Usually because we need to so in order to answer *something*
- But it is important to look back at the big question
- Think that in **every decision** on RQs, concepts, measurements, strategy... you lose some detail

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2. Importance of meta-scientific knowledge

Quantitative approaches for looking at the big picture

1. Robustness of results and incremental science (→ meta-analyses)
2. Generalization of results (→ construct and external validity)

An example of a meta-analysis

- Effect of natural resources (oil, diamonds, etc) on conflict
- What's the causal inference problem here?

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- What's the causal inference problem here?
- But there are options to solve this, right?

An example of a meta-analysis

Do Commodity Price Shocks Cause Armed Conflict? A Meta-Analysis of Natural Experiments

GRAEME BLAIR *University of California, Los Angeles*

DARIN CHRISTENSEN *University of California, Los Angeles*

AARON RUDKIN *University of California, Los Angeles*

Scholars of the resource curse argue that reliance on primary commodities destabilizes governments: price fluctuations generate windfalls or periods of austerity that provoke or intensify civil conflict. Over 350 quantitative studies test this claim, but prominent results point in different directions, making it difficult to discern which results reliably hold across contexts. We conduct a meta-analysis of 46 natural experiments that use difference-in-difference designs to estimate the causal effect of commodity price changes on armed civil conflict. We show that commodity price changes, on average, do not change the likelihood of conflict. However, there are cross-cutting effects by commodity type. In line with theory, we find price increases for labor-intensive agricultural commodities reduce conflict, while increases in the price of oil, a capital-intensive commodity, provoke conflict. We also find that price increases for lootable artisanal minerals provoke conflict. Our meta-analysis consolidates existing evidence, but also highlights opportunities for future research.

- Idea: Commodity **prices** are often **exogenous** to internal conflict

How are meta-analyses done

1. Literature review → 3,300 records

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“the 46 included studies (102 estimates) quantitatively analyze the effect of plausibly exogenous variation in world commodity prices on armed civil conflict by using a generalized difference-in-difference design with unit and time fixed effects”

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4. Perform meta-analysis method

fixed-effects meta-analysis model & random effects m-a model

Theoretical background on natural resources



Theoretical background on natural resources

H1: NR are the 'prize' to be won by fighting (+)

H2: Higher prices change opportunity cost (- for labor-intensive, + for capital-intensive)

H3: Lootable NR can be captured by fighting (+)

TABLE 1. Commodity Classifications and Predicted Effect Direction from Each Hypothesis

Commodity type	Characteristics		Predicted direction		
	Labor-intensive	Lootable	(H1)	(H2)	(H3)
Pooled (average of commodities)	Mix	Mix	+	+/-	+/0
Agriculture	✓		+	-	0
Artisanal minerals	✓	✓	+	-	+
Commercial minerals			+	+	0
Oil & gas		✓	+	+	+/0
Bundle of multiple types	Mix	Mix	+	+/-	+/0

Meta-analysis findings

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Meta-analysis findings

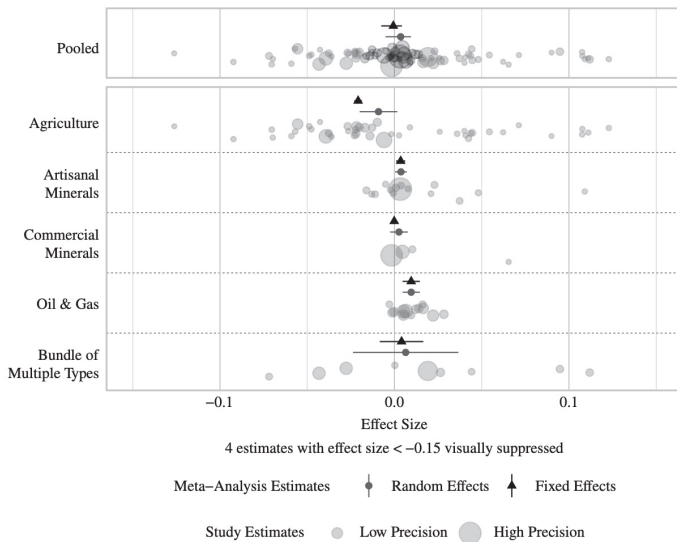
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 - price increases for **agricultural commodities** **reduce** the likelihood of armed conflict (labor-intensive commodities and opportunity cost)

Meta-analysis findings

- **Pooling studies** across commodity types: no effect
- But when **disaggregating by commodity type**: effects coherent with theoretical predictions
 - price increases for **agricultural commodities** **reduce** the likelihood of armed conflict (labor-intensive commodities and opportunity cost)
 - price increases for **oil and gas** **increase** likelihood of conflict (capital-intensive commodity and returns to fighting)

Meta-analysis findings

FIGURE 1. Effects of Commodity Prices on Conflict by Commodity Type



Learning about gaps

- A meta-analysis can be used to identify **gaps**, e.g.
 1. **Spatial or temporal validity**: where and when do findings come from?
 2. **Publication bias**: are only positive findings being published?

FIGURE 2. Evidence Gap Map (Number of Estimates) by Commodity Type and Continent

Agriculture	N = 14	9	3	2	17
Artisanal Minerals	12	1		3	
Commercial Minerals	4				
Oil & Gas	4	2		3	4
Multiple	4				6
	Africa	Asia	N. America	S. America	Multiple

Funnel plots

- Looking for **publication bias**
 - are people only submitting/publishing certain results? (so we'd not have the full picture)

- References [here](#) and [here](#)

Funnel plots

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- **Funnel plot**: scatter plot of reported results
 - X-axis: Effect size
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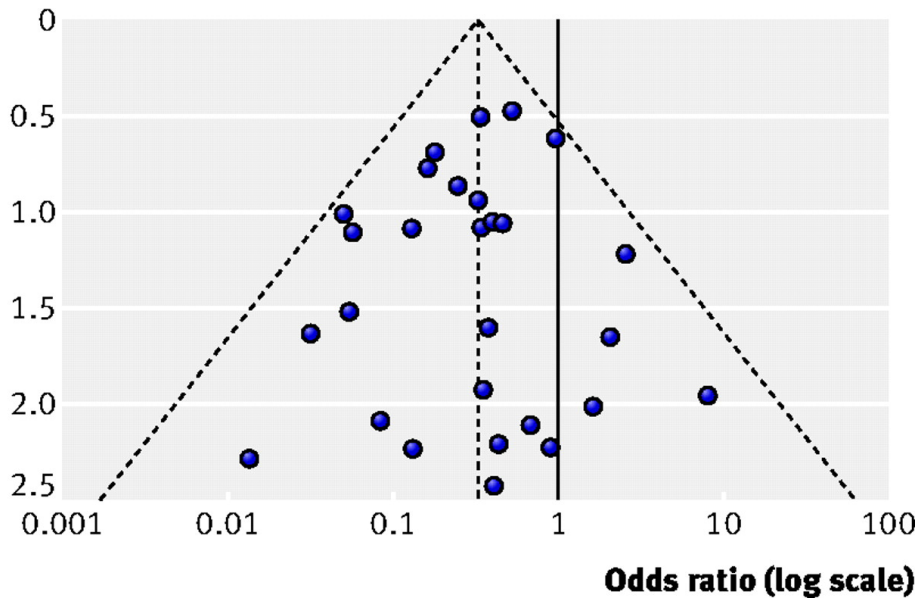
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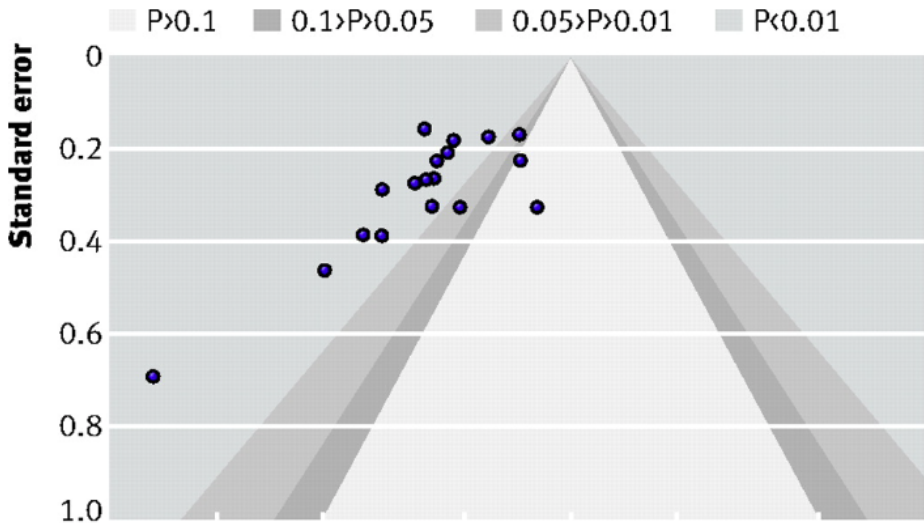
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- We should see a *pyramid*: less precision at the bottom (more spread-out effect sizes), more precision at the top (center at truth)
- If there's publication bias, e.g. we won't see points in the bottom right/left (only positive or negative effects are submitted/published)
- References [here](#) and [here](#)

Standard error





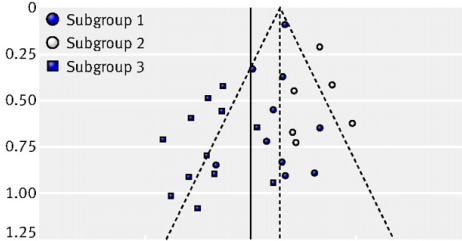
Other cases

- Weird funnel plots could also indicate other things different from publication bias
- Asymmetry could be due to **unknown heterogeneity**

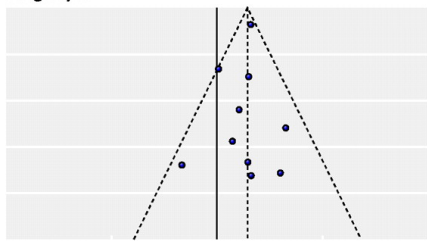
Standard error

All studies

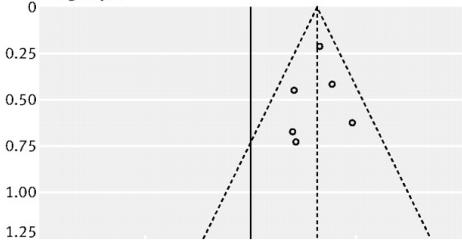
- Subgroup 1
- Subgroup 2
- Subgroup 3



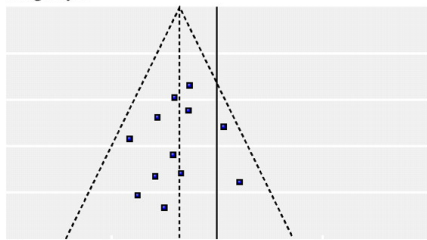
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Subgroup 2



Subgroup 3

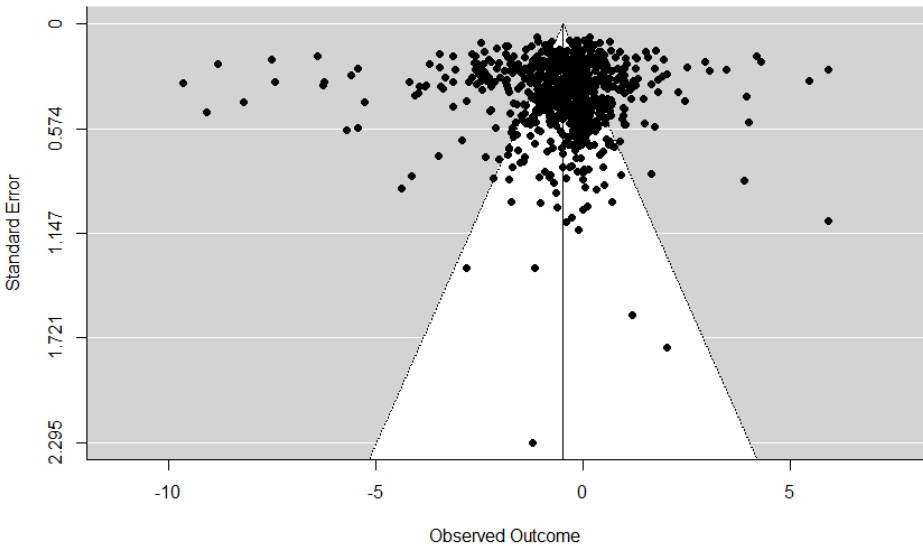


Odds ratio (log scale)

Odds ratio (log scale)

Other cases

- Weird funnel plots could also indicate other things different from publication bias
- Asymmetry could be due to **unknown heterogeneity**
- Or we could observe an **inverted funnel plot**
- That could be the case e.g. if people don't submit 'imprecise' estimates but still can't find the true effect (unaccounted for heterogeneity, as in the natural resources example?)
 - See e.g. [this post](#)



Generalizing results

- Meta-analyses are **not** enough
- There are also formal approaches to understand the validity of studies
- Two key concepts

Generalizing results

- **Construct validity**: are we really measuring what we intent to?
 - is the treatment doing what we theoretically expect it to? and are we measuring the outcome correctly?
- **External validity**: would we get the same results if we replicate this in another context?
 - especially: temporal and spatial validity

Construct validity

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Construct validity

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 - We say we are analyzing the effect of X on Y , but are we really measuring that?
1. Construct validity of the **cause**: bundled treatments, exclusion restriction (assignment only affects through treatment), SUTVA, etc
 2. Construct validity of the **outcome**: similar, making sure the empirical measure mirrors the theoretical construct: thing e.g. about measuring polarization

External validity (Egami and Hartman 2023)

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(it is very unlikely that the effect takes place under universal conditions)
- ‘Effect-generalization’ and the softer ‘sign-generalization’
- Different types of external validity:
 - Populations (X -validity)
 - Treatments (T -validity)
 - Outcomes (Y -validity)
 - Contexts and settings (C -validity)

TABLE 1. Summary of Typology

	Practical concerns (examples)	Causal assumptions (formalization)
X-validity	Convenience samples, survey non-response, attrition	Ignorability of sampling and treatment effect heterogeneity (Assumption 1)
T-validity	Realistic treatments, bundled treatments, difference in implementations	Ignorable treatment variations (Assumption 2)
Y-validity	Behavioral or hypothetical survey outcomes, short- or long-term outcomes	Ignorable outcome variations (Assumption 3)
C-validity	Mechanisms differ across time, geography, political institutions, and so on	Contextual exclusion restriction (Assumption 4)

The special case of *temporal validity*? (Munger 2023)

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- Existing approaches use the idea of treatment $\times$ covariate space to know where and to what extent we can generalize, but under certain conditions, the problem is that that space *changes or expands*
 - Example of studies on dissent and the Arab Spring & importance of smartphones

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 - Example of studies on dissent and the Arab Spring & importance of smartphones
- How do we solve this? Importance of **qualitative studies and theory**

References on external validity

- Egami and Hartman (2023) **Elements of External Validity: Framework, Design, and Analysis**. *APSR* 117(3): 1070–1088.
- Munger (2023) **Temporal validity as meta-science**. *Res&Pol* 10(3).
→ See also this post: **Generalizing Knowledge of Twitter to “X”**.
- Esterling, Brady, & Schwitzgebel (2023) **The Necessity of Construct and External Validity for Generalized Causal Claims: A Critical Review of the Literature on Quantitative Causal Inference**. Preprint.

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Complementary tests

Extra: Synthetic control methods

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Causal methods again, in detail

Re-cap & example

Complementary tests

- Once we have the main analysis, things do not end there
- We need more evidence supporting our argument and increasing credibility of results
- Two interrelated ideas:
 - Testing **additional theoretical implications**
 - Testing the **robustness of analyses**

Additional implications

1. Mechanism

→ How could we obtain evidence support the mechanism we propose?

2. Additional empirical observations

→ if our theory is true, what else should we observe or not?

→ can we rule out *other* explanations/mechanisms?

- These two can also overlap

Robustness tests

1. Alternative specifications

- what happens if we change measurement decisions, model choices, sample, etc?
- testing methodological leverage

2. Placebo tests

- Testing assumptions of the inference strategy
- Looking for relationships where there should be none
- Remember parallel trends assumption in DiD

Types of placebo tests

- Eggers, Tuñón & Dafoe (2023) Placebo Tests for Causal Inference
- Example: Peisakhin & Rozenas on effect of Russian news media and exploiting news TV signal strength

1. Placebo outcome test

- Testing alternative outcomes
- Ex: Russian TV signal strength should *not* predict % Russian speakers or other confounders

2. Placebo treatment test

- Testing alternative treatments
- Ex: Russian *sports* TV signal should not predict voting

3. Placebo population test

- Looking at a population where there should be no effect
- Ex: Ukrainians that watch satellite TV (signal shouldn't matter)

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Synthetic control methods

- Imagine combining diff-in-diff with matching
 - treatments goes into effect at one time, and only affects one group
 - you can use pre-T data to make control and treated comparable
 - you estimate effect from post-T divergence

The Economic Costs of Conflict: A Case Study of the Basque Country

By ALBERTO ABADIE AND JAVIER GARDEAZABAL*

This article investigates the economic effects of conflict, using the terrorist conflict in the Basque Country as a case study. We find that, after the outbreak of terrorism in the late 1960's, per capita GDP in the Basque Country declined about 10 percentage points relative to a synthetic control region without terrorism. In addition, we use the 1998–1999 truce as a natural experiment. We find that stocks of firms with a significant part of their business in the Basque Country showed a positive relative performance when truce became credible, and a negative relative performance at the end of the cease-fire. (JEL D74, G14, P16)

American Economic Review, 2003

Original study

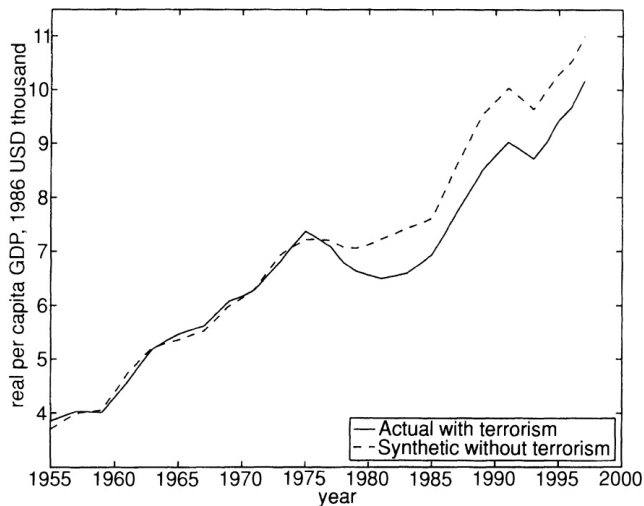


FIGURE 1. PER CAPITA GDP FOR THE BASQUE COUNTRY

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- By the way: complementary tests of additional implications

Reference example

- A Abadie, A Diamond & J Hainmueller (2010) **Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program.** *Journal of the American Statistical Association.*

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Questions about final essay

- Review something?
- Format and guidelines
- Group vs individual essays
- Logistics: deadline, where to submit

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Re-cap & example

Controlling in regression

- We probably already know this
- We've seen what it means controlling: just adjusting for the variation already predicted by the other variables
- One limitation though: we need to *observe* those variables

Controlling

- We are going to see two further methods of controlling widely used:
 - Matching
 - Fixed effects

Matching

- Adding control variables is not the only way to control / close back doors
- Imagine we have the following model:
 - $Z \rightarrow X \rightarrow Y \leftarrow Z$
 - Where Z is whether someone is retired or not
- If we select a sample of *only* retired people, we are closing that back door
 - $(X \leftarrow Z \rightarrow Y)$
- Matching is something like this, it's basically about creating groups of comparison where Z (which can be several variables) does not vary

How matching works

- We have a treatment group and a control group (so: *binary* treatment variables)
- The main idea: give different *weights* to treated and control observations, so we eliminate
- We get these weights by using one or more *matching variables* (i.e. confounding variables)

How matching works

- Imagine we have:
 - Treatment: get some specific skill training
 - Outcome: get a job afterwards
 - Confounding variable: gender
- Control group: 80 men and 20 women
 - 75% of men get a job, 60% of women do
- Treatment group: 500 men and 500 women
 - 70% of men get a job, 55% of women do
- Comparing within each group, we know the treatment effect is a 5% increase in the odds of getting a job ($70 \rightarrow 75$, $55 \rightarrow 60$)
- But if we do the global comparison, it's almost 10%
 - $60 \text{ men} + 12 \text{ women out of } 100 = 72/100 = 72\%$, vs $(350 + 275)/1000 = 62.5\%$, so a difference of 9.5 points

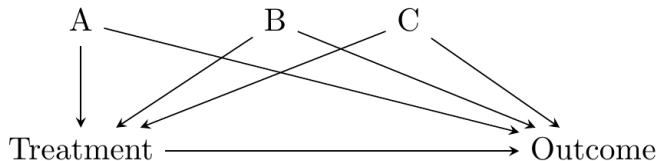
How matching works

- The problem is we have 4 times more men in the treated group than women, whereas we have equal proportion on the control group
- So we'll weight the control observations by gender, giving *more* weight to the men observations, so it looks more similar to the treatment group
- $(4 * 350 + 1 * 275) / (4 * 500 + 1 * 500) = 67\%$
- Now, the unweighted difference in the treatment group was 72%, and now the weighted difference in the control group is 67%
- The difference is 5 points, the same as the within-group calculation we did before

Two approaches to matching

1. Distance matching

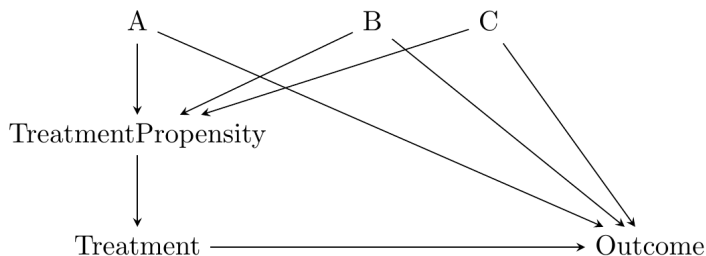
- We want to create a dataset where treatment and control observations have similar values (distance) in the confounding variables
- If, say, our confounding variable is income, we'll pick control observations that have a similar value on income to each of the treatment observations
- <https://nickchk.com/causalgraphs.html>



Two approaches to matching

2. Propensity score matching

- We want to account for the differential likelihood in getting into treatment depending on the value of the confounding variables
- We estimate the probability of getting into treatment, usually by doing a regression where the outcome is the treatment and the right-hand variables are the confounders
- We control for the propensity score matching, or select based on it (or both)



Two approaches to matching: propensity score

```
df = data.frame(  
  treat = c(rep(1, 100), rep(0, 1000)),  
  gender = c(rep("M", 80), rep("F", 20), rep(c("M","F"), each = 500)),  
  y = NA  
)  
  
df$y[df$treat == 1 & df$gender == "M"] = rbinom(80, 1, 0.75)  
df$y[df$treat == 1 & df$gender == "F"] = rbinom(20, 1, 0.6)  
df$y[df$treat == 0 & df$gender == "M"] = rbinom(500, 1, 0.7)  
df$y[df$treat == 0 & df$gender == "F"] = rbinom(500, 1, 0.55)  
  
m1 = glm(y ~ treat, data = df)  
m2 = glm(y ~ treat + gender, data = df)  
modelsummary(list(m1, m2))  
  
ps = glm(treat ~ gender, data = df)  
df$propensity_score = predict(ps, newdata = df)  
m3 = glm(y ~ treat + propensity_score, data = df)  
modelsummary(list(m1, m2, m3))
```

Two approaches to matching: propensity score

	Model 1	Model 2	Model 3
(Intercept)	0.623	0.550	0.493
	(0.015)	(0.021)	(0.030)
treat	0.007	-0.037	-0.037
	(0.051)	(0.051)	(0.051)
genderM		0.147	
		(0.029)	
propensity_score			1.478
			(0.296)
Num.Obs.	1100	1100	1100

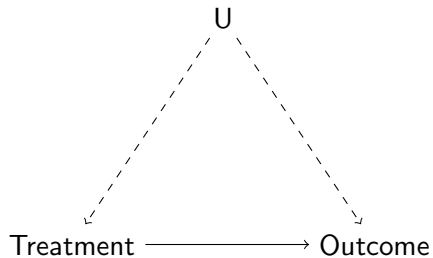
Matching vs regression

- Matching and regression are complementary approaches
- e.g. regression doesn't waste any information, but has a linearity assumption
- It's usual to use both at the same time

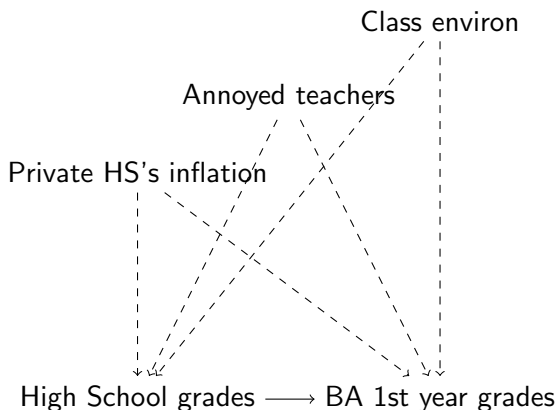
Fixed effects

- The problem with covariate adjustment, regardless of whether we use regression or matching, is that we need to *observe* those variables
- But another strategy when we have unobserved confounders is to try within-group comparisons, which will work when the unobserved variance is constant within some group
- For example, imagine cases when our U variable is:
 - *Country history*, in a cross-national analysis
 - *City of origin*, in an individual-level analysis
 - *Individual background*, in a panel survey analysis
 - *Company effects*, if we look at the effects of English courses on internal promotion using individual data from many different companies
 - etc

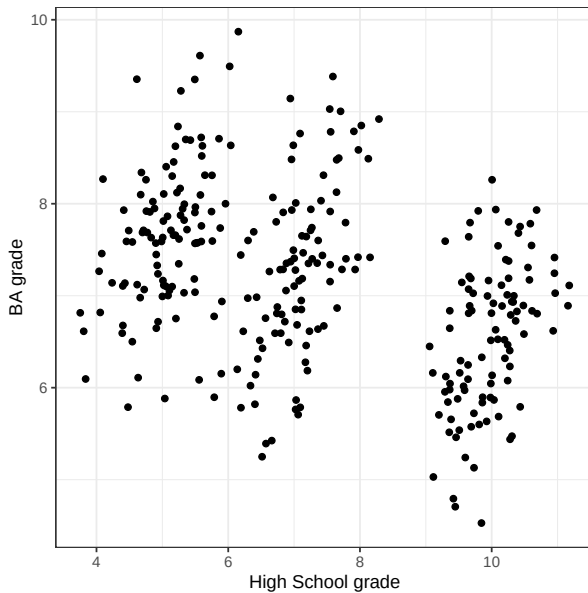
Fixed effects - when do we use them?



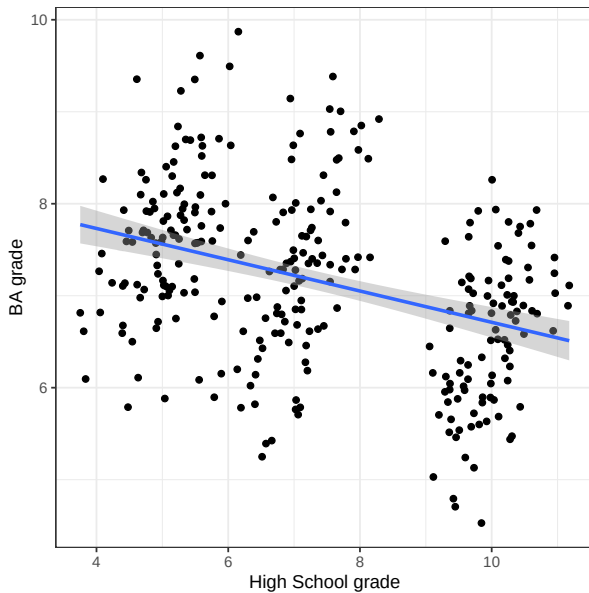
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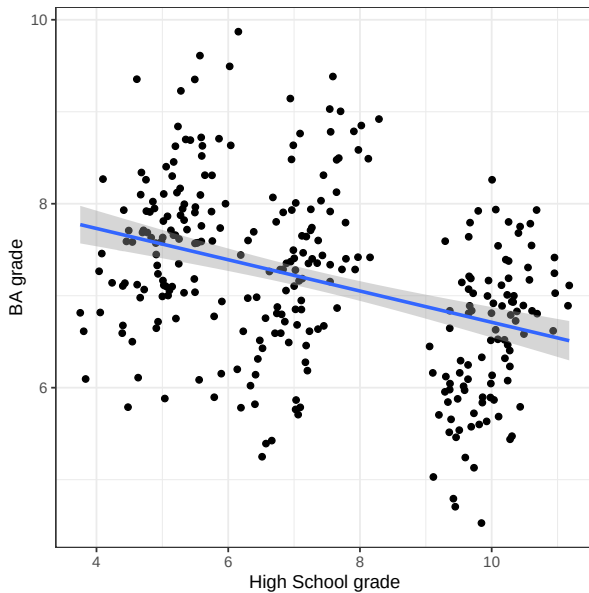
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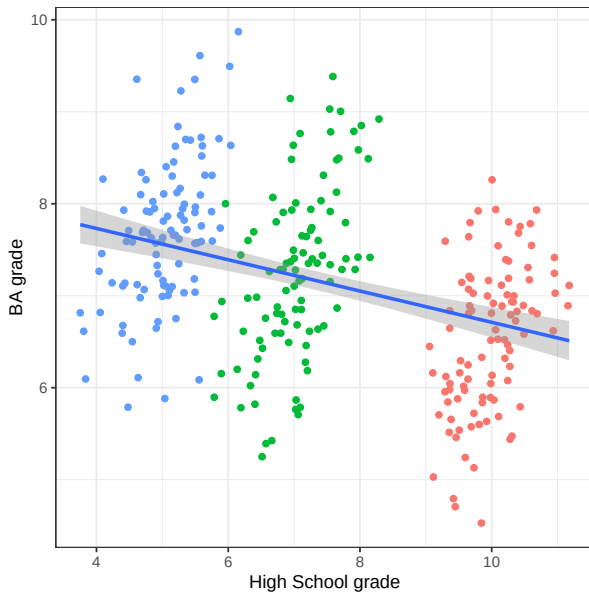
Fixed effects



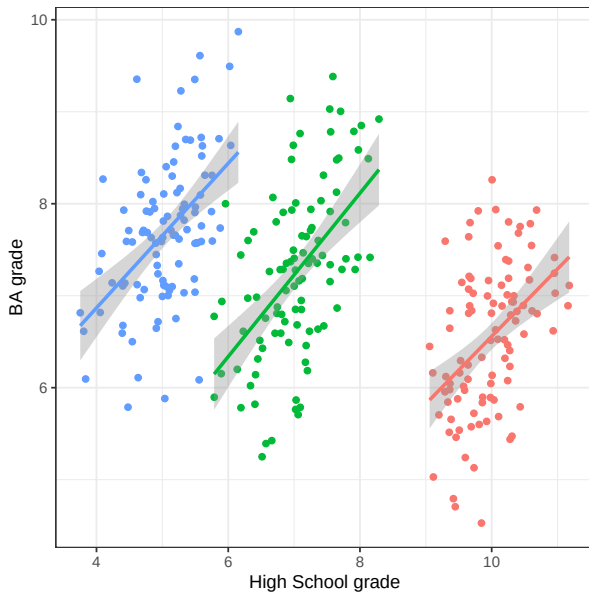
Fixed effects



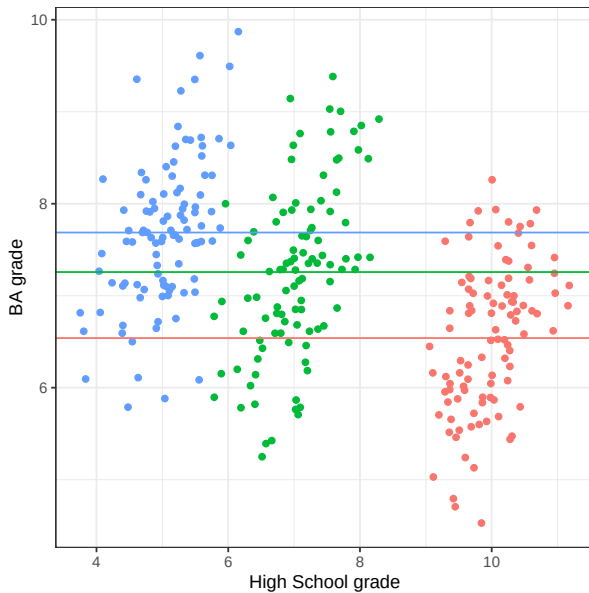
Fixed effects



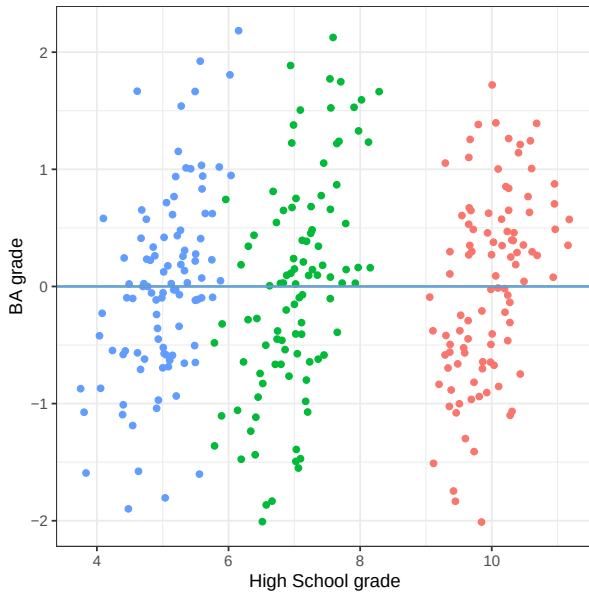
Fixed effects



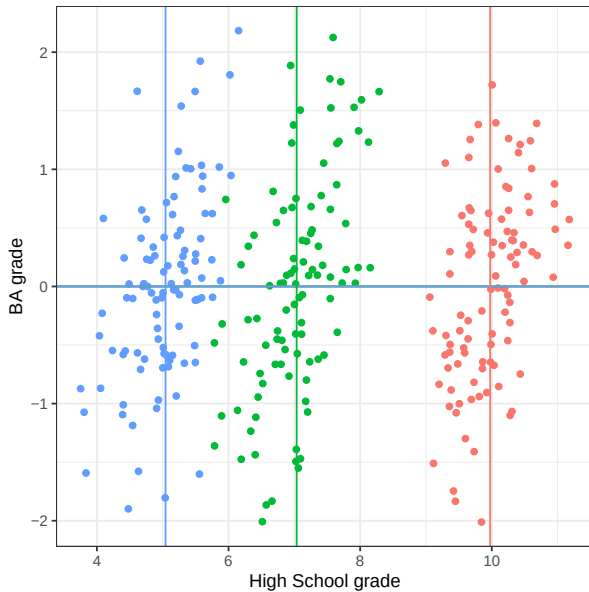
Fixed effects



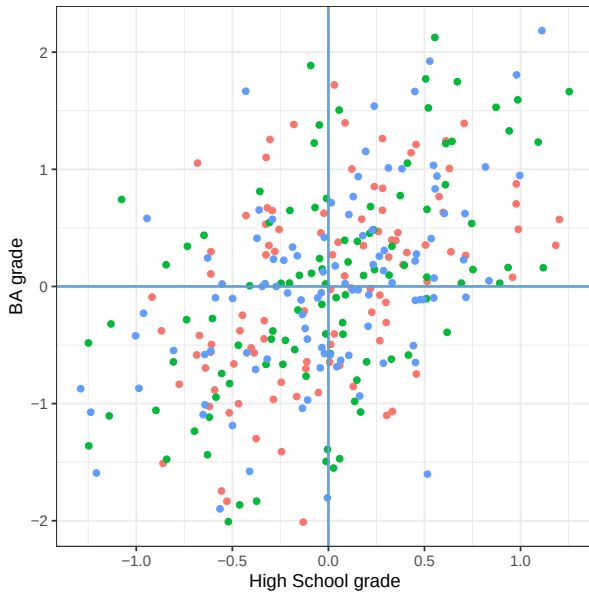
Fixed effects



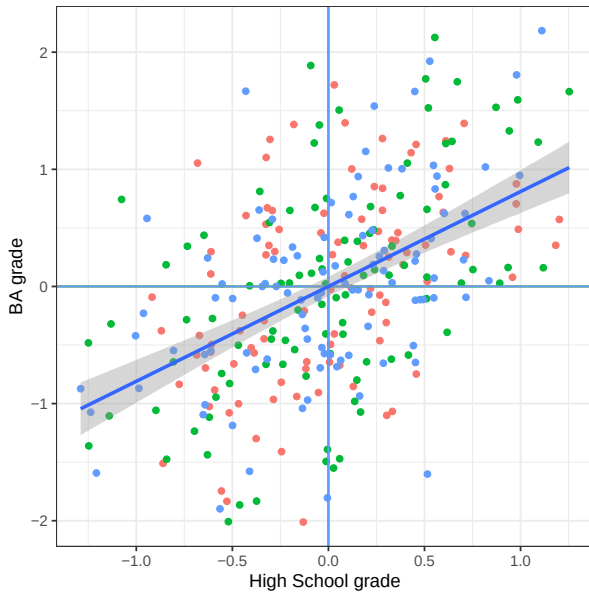
Fixed effects



Fixed effects



Fixed effects



Fixed effects and regression

```
# Simulate data: 300 students from three high schools,  
# normally distributed grades  
df = data.frame(high_school = rep(c("A", "B", "C"), each = 100)) %>%  
  group_by(high_school) %>%  
  mutate(hs_grade = rnorm(100, 7, 0.5)) %>%  
  ungroup()  
  
# But school A inflated grades a lot, and C was notoriously difficult  
df$hs_grade[df$high_school == "A"] = df$hs_grade[df$high_school == "A"] + 3  
df$hs_grade[df$high_school == "C"] = df$hs_grade[df$high_school == "C"] - 2  
  
# First-year BA grades are a function of the HS grade +/- the school inflation  
df$ba_grade = 2 + 0.75 * df$hs_grade + rnorm(300, 0, 0.75)  
df$ba_grade[df$high_school == "A"] = df$ba_grade[df$high_school == "A"] - 3  
df$ba_grade[df$high_school == "C"] = df$ba_grade[df$high_school == "C"] + 2
```

Our true causal model

Fixed effects and regression

```
Call:
lm(formula = ba_grade ~ hs_grade, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-2.42897 -0.57050 -0.05923  0.66227  2.76385

Coefficients:
              Estimate Std. Error t value      Pr(>|t|)
(Intercept)  8.70808     0.19412   44.86 < 0.0000000000000002 ***
hs_grade    -0.22059     0.02544   -8.67 0.000000000000000283 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9061 on 298 degrees of freedom
Multiple R-squared:  0.2014, Adjusted R-squared:  0.1988
F-statistic: 75.17 on 1 and 298 DF, p-value: 0.0000000000000002826
```


Fixed effects and regression

```
Call:
lm(formula = ba_grade ~ hs_grade + factor(high_school), data = df)

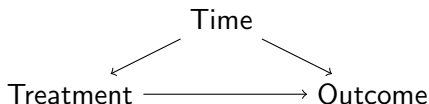
Residuals:
    Min       1Q   Median       3Q      Max
-2.0124 -0.5464  0.0042  0.4984  2.2539

Coefficients:
                Estimate Std. Error t value      Pr(>|t|)
(Intercept)      -1.68390    0.98549   -1.709    0.0886 .
hs_grade           0.80803    0.09871    8.186 0.00000000000000811 ***
factor(high_school)B  3.23797    0.30984   10.451 < 0.0000000000000002 ***
factor(high_school)C  5.26454    0.49388   10.659 < 0.0000000000000002 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7719 on 296 degrees of freedom
Multiple R-squared:  0.4243, Adjusted R-squared:  0.4185
F-statistic: 72.73 on 3 and 296 DF, p-value: < 0.00000000000000022
```

Difference-in-differences

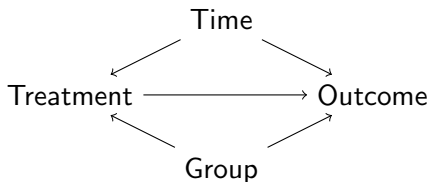
- Treatments usually occur at a particular moment in time, e.g.:
 - Minimum wage increase
 - Terrorist attack
 - Influx of refugees
 - ...
- In those cases, if we have before & after observations, we have something like this:



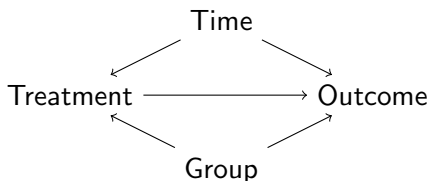
- The **problem** is that **all** the variation goes through time, so if we close that back door, we're left with nothing

Difference-in-differences

- So one strategy we can use is to bring additional group that is *not treated* and for which we also have before/after observations
 - Minimum wage increase: maybe those earning above MW?
 - Influx of refugees: other countries? regions far from the border?
 - Terrorist attack: do we have a control (untreated) group?
 - ...

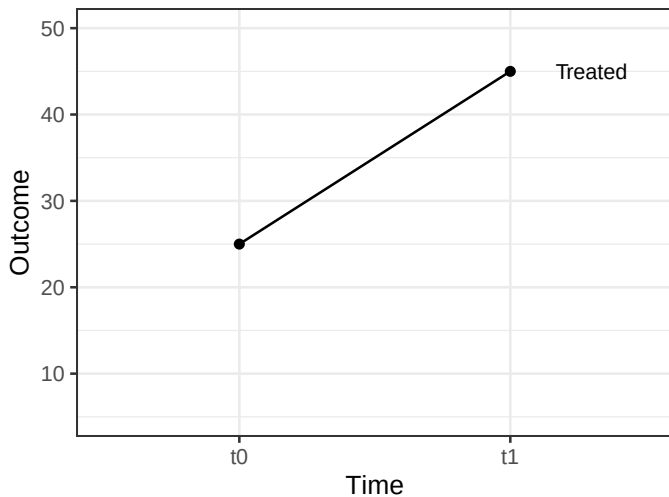


Difference-in-differences

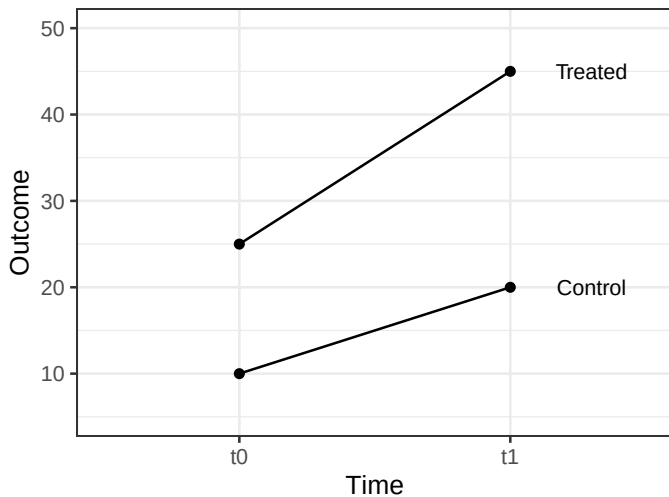


- We can compare changes across time *within* the treated and control groups (closing the back door through group)
- Compare within-group variation between treated and control (since time affects both 'within-variations' the same way, we are closing the other back door through time)

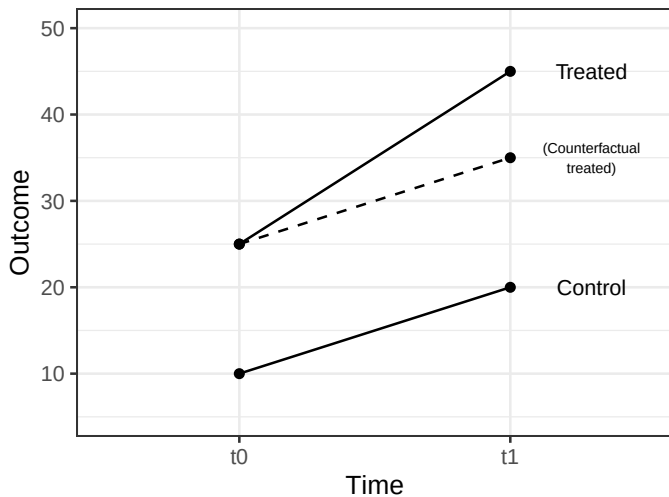
Difference-in-differences



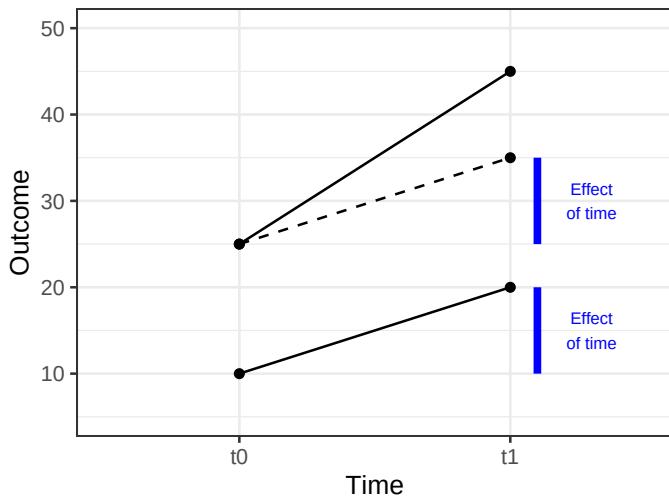
Difference-in-differences



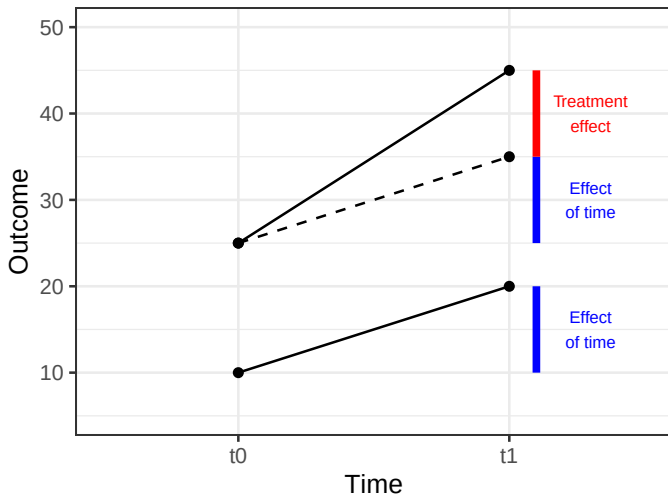
Difference-in-differences



Difference-in-differences



Difference-in-differences



Difference-in-differences: Cholera in London



- Probably first use of DiD and natural experiments?
- Lambeth Company moved water intake upriver in 1952, Southwark & Vauxhall Company still got it from downstream

Difference-in-differences: Cholera in London

2261	2458	Southwk. & Vauxhall.
3905	2547	Both Companies.
162	37	Lambeth Company.

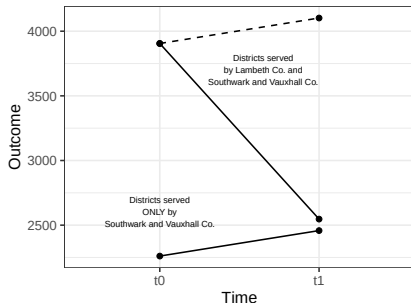


TABLE XII.

Sub-Districts.	Deaths from Cholera in 1849.	Deaths from Cholera in 1854.	Water Supply.
St. Saviour, Southwark .	283	371	Southwark & Vauxhall Company only.
St. Olave .	157	161	
St. John, Horsleydown .	192	148	
St. James, Bermondsey .	240	362	
St. Mary Magdalen .	259	244	
Leather Market .	226	237	
Rotherhithe* .	332	282	
Wandsworth .	97	59	
Battersea .	111	171	
Putney .	8	9	
Camberwell .	235	240	
Peckham .	92	174	
Christchurch, Southwark	256	113	Lambeth Company, and Southwark and Vauxhall Comp'y.
Kent Road .	267	174	
Borough Road .	312	270	
London Road .	257	93	
Trinity, Newington .	318	210	
St. Peter, Walworth .	446	388	
St. Mary, Newington .	143	92	
Waterloo Road (1st) .	193	58	
Waterloo Road (2nd) .	243	117	
Lambeth Church (1st) .	215	49	
Lambeth Church (2nd) .	544	193	
Kennington (1st) .	187	303	
Kennington (2nd) .	153	142	
Brixton .	81	48	
Clapham .	114	165	
St. George, Camberwell	176	132	
Norwood .	2	10	Lambeth Company only.
Streatham .	154	15	
Dulwich .	1	—	
Sydenham .	5	12	
First 12 sub-districts .	2261	2458	Southwk. & Vauxhall.
Next 16 sub-districts .	3905	2547	Both Companies.

Difference-in-differences

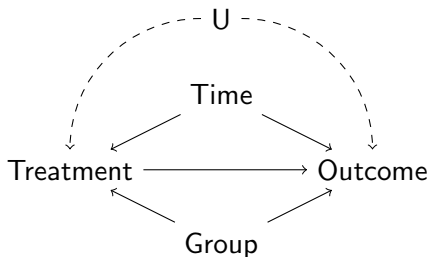
- You can estimate this effect just with group means
- But it is often easier to use regression, also because you can include controls:

$$Y_{it} = \beta_0 + \beta_1 Treated_i + \beta_2 After_t + \beta_3 (Treated_i \times After_t) + \beta^\top \mathbf{x}_i + \epsilon_{it} \quad (1)$$

- But why do we need all this?

Difference-in-differences

- Because DiD identification **depends** on the assumption that the **control group is a good counterfactual to the treated group**

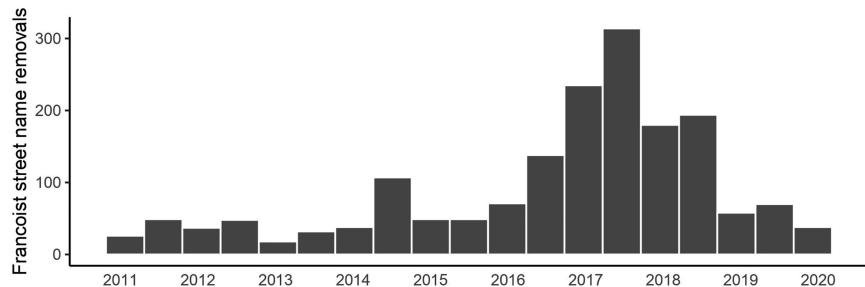


- One way to test this is checking if the **parallel trends assumption** holds (we need data further back in time)

DiD example

- What is the effect of symbolic TJ policies?
- journals.sagepub.com/doi/full/10.1177/20531680211058550

DiD example



DiD example

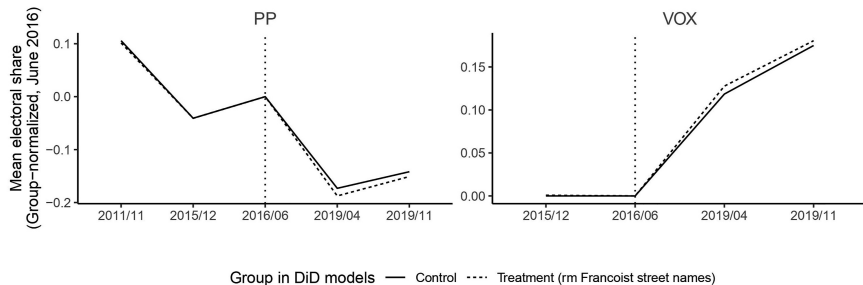
Francoist names	Removed Francoist names, 2016–2018?	
In June 2016?	No	Yes
No	6455	0
	(100%)	(0%)
Yes	1184	454
	(72%)	(28%)
<i>Note: Row percentages. Changes in 2016–2018 refer to the period between 30/06/2016 and 31/12/2018.</i>		

DiD example

Table 2. Mean electoral share in sample.

	June 2016			April 2019			
Party	Control	Treated	Δ	Control	Treated	Δ	$\Delta_{2019}-\Delta_{2016}$
Vox	0.21	0.21	0	12.54	13.28	0.74	0.74
PP	41.22	46.77	5.55	23.83	27.68	3.85	-1.7
PSOE	29.13	28.01	-1.12	33.38	32.03	-1.35	-0.23

DiD example



Regression discontinuity

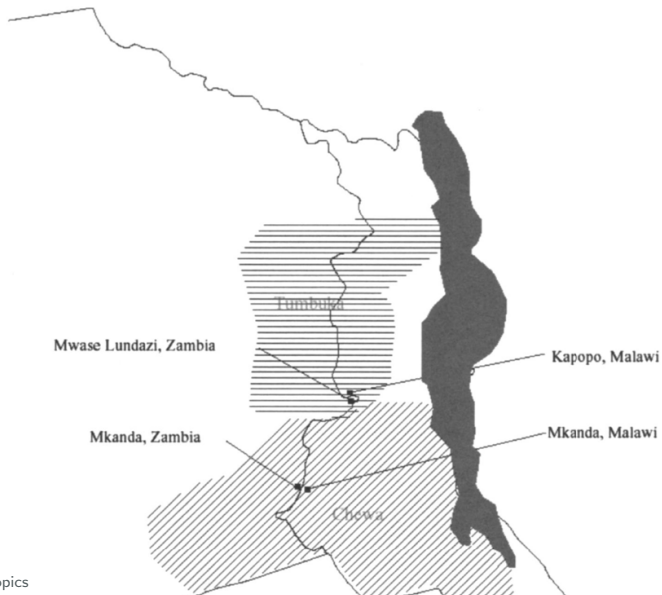
The Political Salience of Cultural Difference: Why Chewas and Tumbukas Are Allies in Zambia and Adversaries in Malawi

DANIEL N. POSNER *University of California, Los Angeles*

This paper explores the conditions under which cultural cleavages become politically salient. It does so by taking advantage of the natural experiment afforded by the division of the Chewa and Tumbuka peoples by the border between Zambia and Malawi. I document that, while the objective cultural differences between Chewas and Tumbukas on both sides of the border are identical, the political salience of the division between these communities is altogether different. I argue that this difference stems from the different sizes of the Chewa and Tumbuka communities in each country relative to each country's national political arena. In Malawi, Chewas and Tumbukas are each large groups vis-à-vis the country as a whole and, thus, serve as viable bases for political coalition-building. In Zambia, Chewas and Tumbukas are small relative to the country as a whole and, thus, not useful to mobilize as bases of political support. The analysis suggests that the political salience of a cultural cleavage depends not on the nature of the cleavage itself (since it is identical in both countries) but on the sizes of the groups it defines and whether or not they will be useful vehicles for political competition.

Regression discontinuity

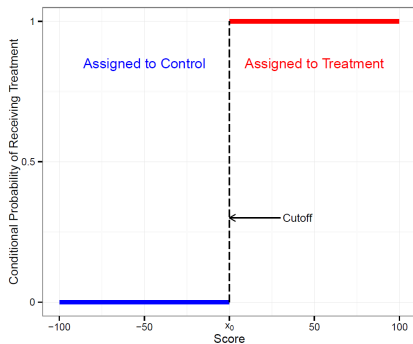
FIGURE 1. Research Sites



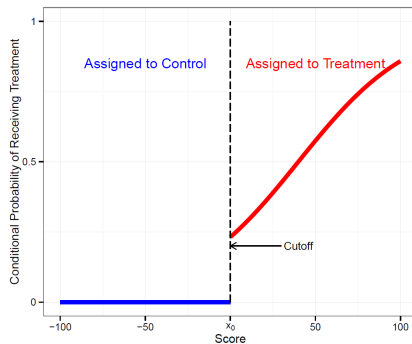
Regression discontinuity

- RDD works well when assignment into treatment depends on a cutoff along a **running variable**
 - Do incumbent politicians have an electoral advantage? (vote share)
 - What is the effect of being drafted into the military? (birth year)
 - Effect of national policies in ethnic identification in Africa? (distance to colonial borders)
- This is the source of the exogenous variation (or if you will, the natural experiment):
 - Although many variables confound the relationships between X and Y , nothing should be too different *around the cutoff* between treatment and control groups (local randomization assumption)
 - Sometimes you look at different *bandwidths* to check this

Figure 1: Conditional Probability of Receiving Treatment in Sharp vs. Fuzzy RD Designs



(a) Sharp RD

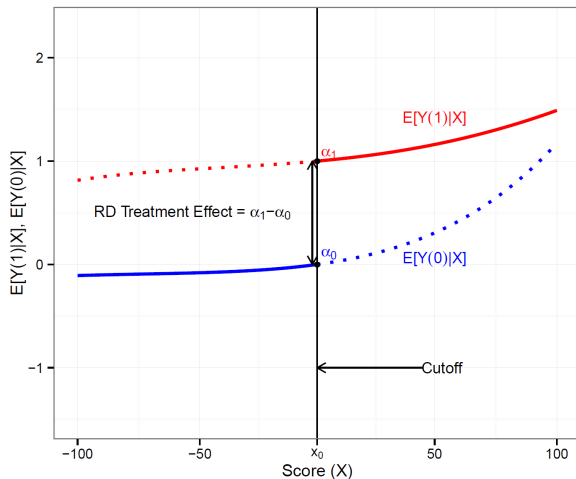


(b) Fuzzy RD (One-Sided)

<https://bookdown.org/paul/applied-causal-analysis/rddbasics2.html>

RDD

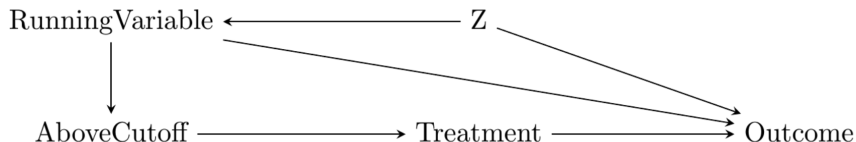
Figure 2: RD Treatment Effect in Sharp RD Design



<https://bookdown.org/paul/applied-causal-analysis/rddbasics3.html>

RDD

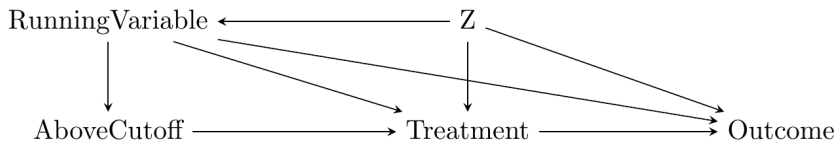
- Underlying assumption: other confounders also vary along the running variable, but are independent to the *jump*



Huntington-Klein, *The Effect*, p.508

RDD

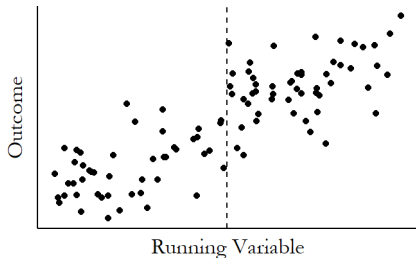
- Underlying assumption: other confounders also vary along the running variable, but are independent to the *jump*
- Even if in a *fuzzy* design



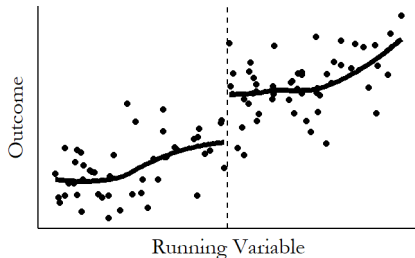
Huntington-Klein, *The Effect*, p.508

RDD implementation

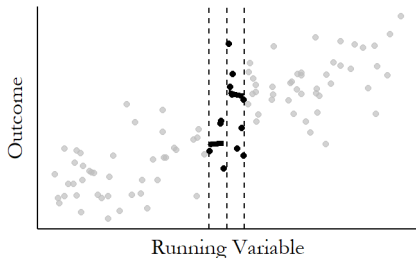
(a) Raw Data



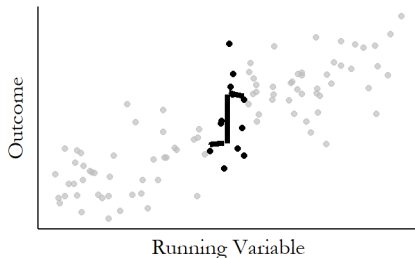
(b) Predict Values Near the Cutoff



(c) Pick a Bandwidth

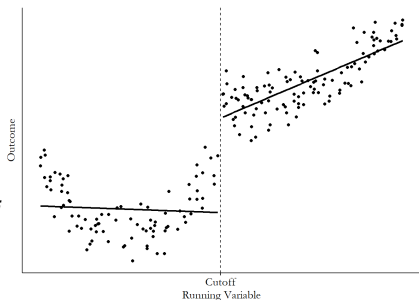


(d) Estimate Jump at the Cutoff



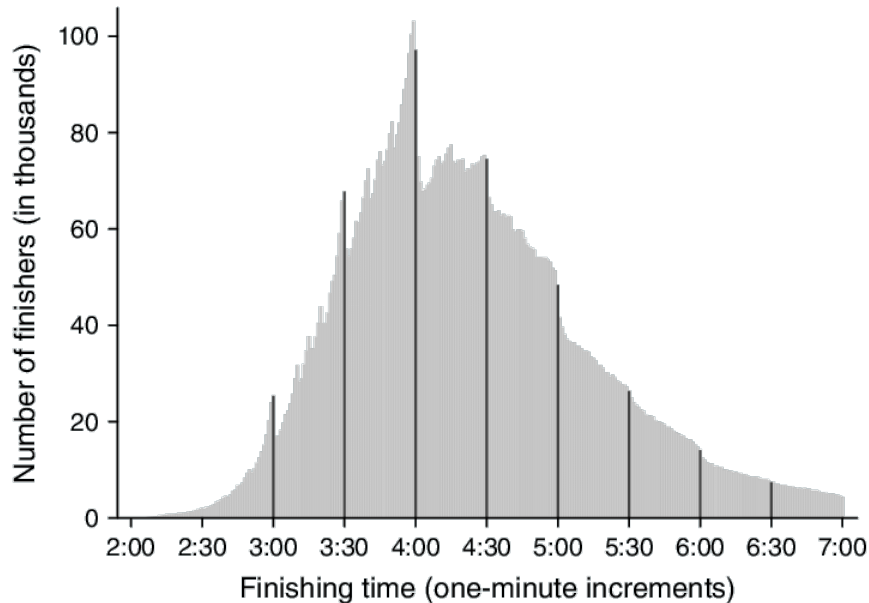
RDD and regression

$$Y = \beta_0 + \beta_1 \text{Distance} + \beta_2 \text{Treated} + \beta_3 (\text{Treated} \times \text{Distance}) + \beta^\top \mathbf{x}_i \quad (2)$$



- But there's no need to use linear regression, other methods available as well

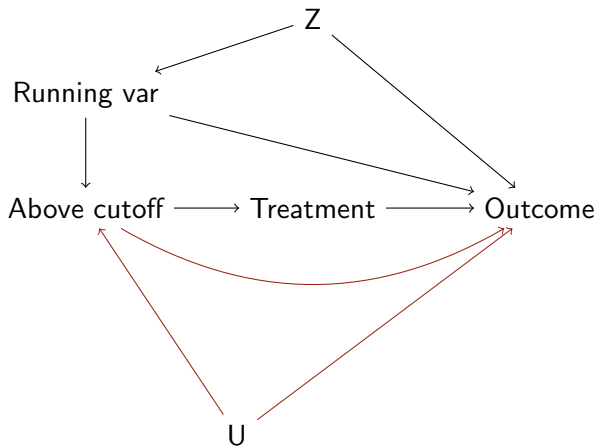
Threats to RDD: precise sorting



Threats to RDD: cutoff ← outcome



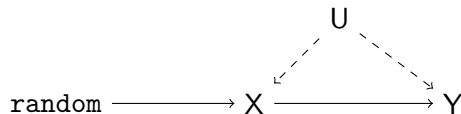
Threats to RDD



Some extensions & combinations

- DiD with multiple treatment periods (units being treated at different times)
- Matched DiD
- Difference-in-discontinuities

Instrumental variables



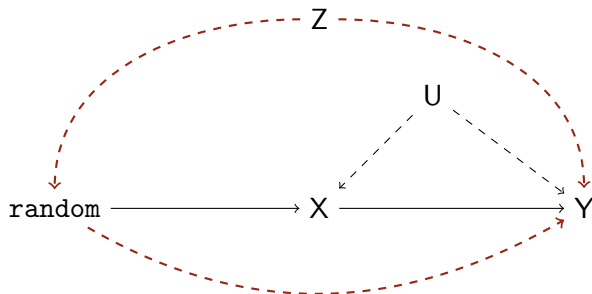
Instrumental variables

- Find an exogenous source of variation in the treatment variable
- Isolate that variation and use it to identify the causal effect

Instrumental variables

- Find an exogenous source of variation in the treatment variable
- Isolate that variation and use it to identify the causal effect
- Assumptions:
 - **Relevance**: the instrument explains at least some part of the treatment variable
 - **Validity** or **exclusion restriction**: no back door paths between the instrument and the outcome

Instrumental variables



Economic Shocks and Civil Conflict: An Instrumental Variables Approach

Edward Miguel

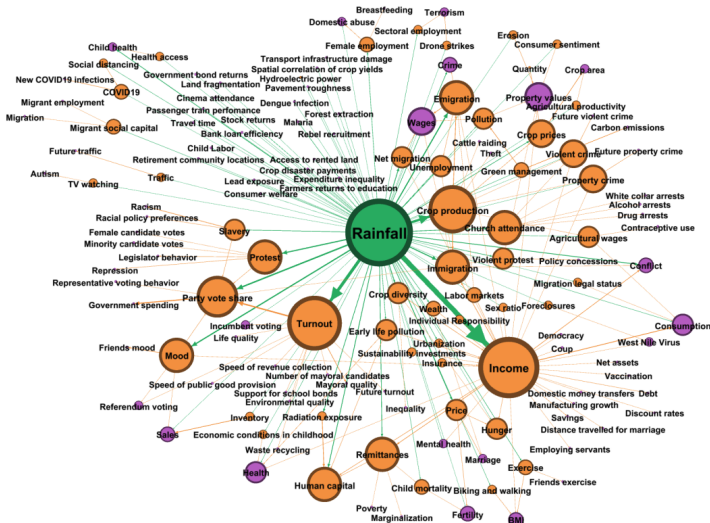
University of California, Berkeley and National Bureau of Economic Research

Shanker Satyanath and Ernest Sergenti

New York University

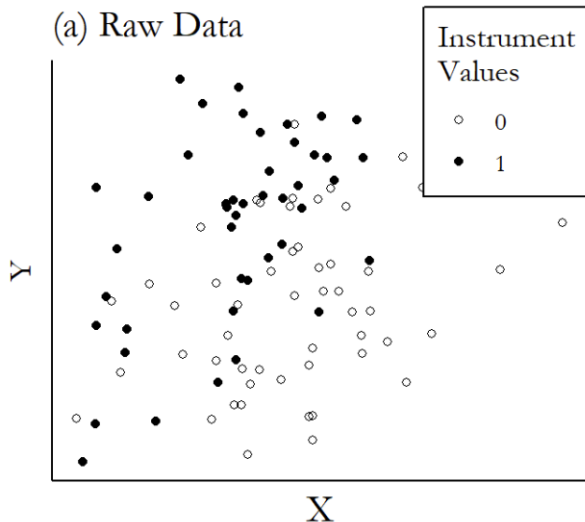
Estimating the impact of economic conditions on the likelihood of civil conflict is difficult because of endogeneity and omitted variable bias. We use rainfall variation as an instrumental variable for economic growth in 41 African countries during 1981–99. Growth is strongly negatively related to civil conflict: a negative growth shock of five percentage points increases the likelihood of conflict by one-half the following year. We attempt to rule out other channels through which rainfall may affect conflict. Surprisingly, the impact of growth shocks

IV threats



Jonathan Mellon (2022) Rain, Rain, Go Away: 192 Potential Exclusion-Restriction

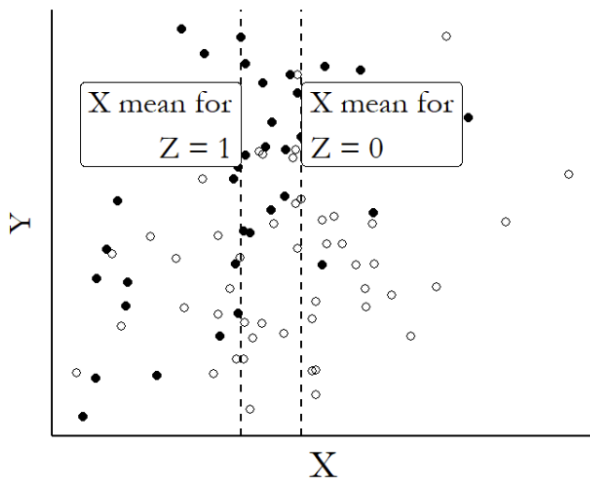
How does IV work?



(Huntington-Klein, *The Effect*, p 472)

How does IV work?

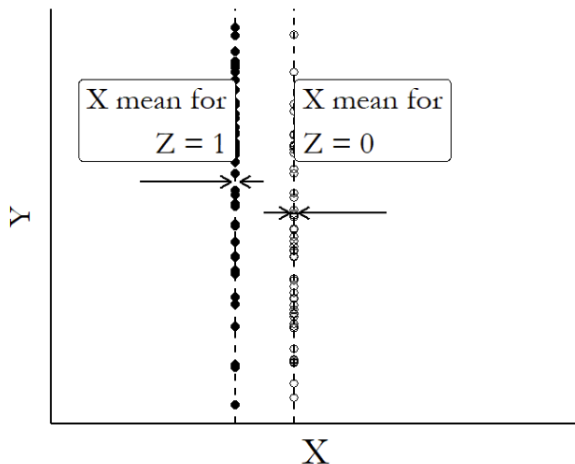
(b) Predict X with Z



(Huntington-Klein, *The Effect*, p 472)

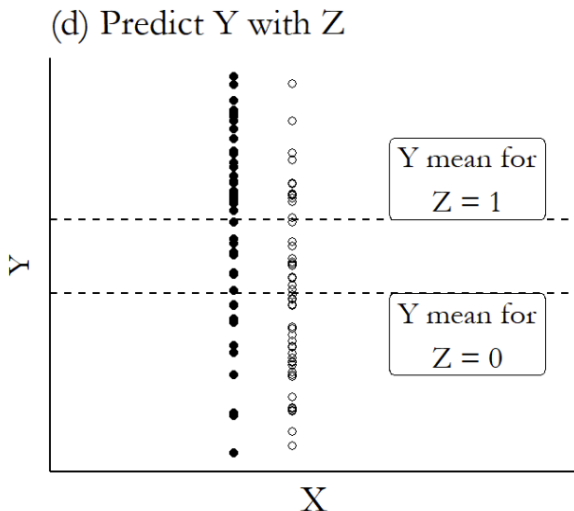
How does IV work?

(c) Only Use Predicted Variation



(Huntington-Klein, *The Effect*, p 472)

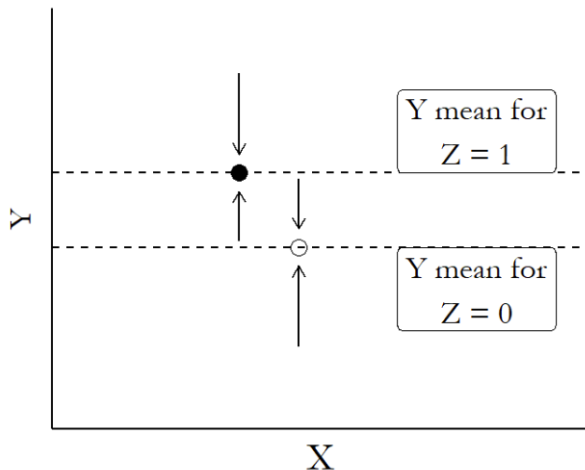
How does IV work?



(Huntington-Klein, *The Effect*, p 472)

How does IV work?

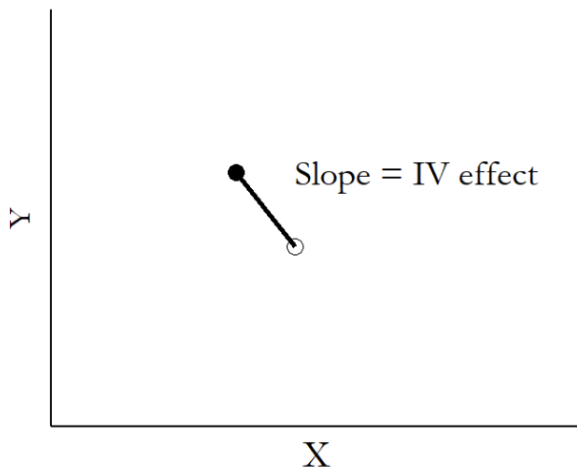
(e) Only Use Predicted Variation



(Huntington-Klein, *The Effect*, p 472)

How does IV work?

(f) Relate Predicted Y to Predicted X



(Huntington-Klein, *The Effect*, p 472)

How does IV work?

- Usually: two-stage least squares, or **2SLS**
1. Run a 'first-stage' regression to predict the treatment with the instrument
 2. Use the predicted values to predict the outcome in the 'second-stage'

Roadmap

Review

Generalizing results

Complementary tests

Extra: Synthetic control methods

Questions

Causal methods again, in detail

Re-cap & example

Re-cap and final essay

- Let's walk through a couple examples:
 1. Rise of protest movements
 2. Violence against civilians and radicalization

E *El Gobierno quiere alejar un nuevo 15-M de las calles*

Las Juventudes Socialistas “serán combativas” sobre vivienda en el congreso del PSOE



La voces de la manifestación



ANABEL DÍEZ

Madrid - 14 OCT 2024 - 05:40 CEST



91

Que el 13-O no se convierta en otro 15-M. Ese es el objetivo del Gobierno. Hace casi tres lustros, un movimiento de indignación, dentro y fuera de España, cristalizó un 15 de mayo de 2011 en el

How the 2003 US-led invasion changed Iraq forever

Fifteen years after the US and its allies invaded Iraq on March 20, 2003, the country is still entrenched in a cycle of sectarian violence and rampant corruption.

[Shatha Al Juburi](#)

9 April 2018



Read more

Get our weekly email

Re-cap and final essay

1. Problem/topic
→ unique to group, general issue
2. Stories, arguments about mechanisms
3. Research question
4. Proper theory, concepts and operationalization
5. Measurement, unit of analyses, data sources, etc
6. Inference strategy
7. Results & interpretation

Re-cap and final essay

1. Problem/topic
2. Stories, arguments about mechanisms
 - testing an argument, testing 2 against each other
3. Research question
4. Proper theory, concepts and operationalization
5. Measurement, unit of analyses, data sources, etc
6. Inference strategy
7. Results & interpretation

Re-cap and final essay

1. Problem/topic
2. Stories, arguments about mechanisms
3. Research question
 - operationalize the story you have in mind
4. Proper theory, concepts and operationalization
5. Measurement, unit of analyses, data sources, etc
6. Inference strategy
7. Results & interpretation

Re-cap and final essay

1. Problem/topic
2. Stories, arguments about mechanisms
3. Research question
4. Proper theory, concepts and operationalization
 - develop a proper argument with view to empirical test
5. Measurement, unit of analyses, data sources, etc
6. Inference strategy
7. Results & interpretation

Re-cap and final essay

1. Problem/topic
2. Stories, arguments about mechanisms
3. Research question
4. Proper theory, concepts and operationalization
5. Measurement, unit of analyses, data sources, etc
→ ideally think about more than one strategy
6. Inference strategy
7. Results & interpretation

Re-cap and final essay

1. Problem/topic
2. Stories, arguments about mechanisms
3. Research question
4. Proper theory, concepts and operationalization
5. Measurement, unit of analyses, data sources, etc
6. Inference strategy
 - depends on previous steps: one per 'arm' or multiple strategies
7. Results & interpretation

Re-cap and final essay

1. Problem/topic
2. Stories, arguments about mechanisms
3. Research question
4. Proper theory, concepts and operationalization
5. Measurement, unit of analyses, data sources, etc
6. Inference strategy
7. Results & interpretation
 - what do we really learn about the initial topic? next steps?

Re-cap and final essay

1. Problem/topic
 2. Stories, arguments about mechanisms
 3. Research question
 4. Proper theory, concepts and operationalization
 5. Measurement, unit of analyses, data sources, etc
 6. Inference strategy
 7. Results & interpretation
- **General guideline:** maybe it's not realistic for you right now, but it needs to be feasible (data collection, etc)